Hector Kohler

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github

Research Experience

2027	Assistant professor (tenure track) at Télécom Paris part of Institut Polytecthnique de Paris. Member of the DIG team in the LTCI lab. Funded by a *Hi! Pairs* chair on Reinforcement Learning
2026	Postdoc researcher on reinforcement learning and graphs, LIX, École Polytechnique, (Advisors: Dr. J. Lutzeyer and Pr. J. Read)
2025	Visiting researcher designing new RL algorithms and evaluation tools, RLAI Lab, University of Alberta, (Advisor: Dr. A. White)
2022-2025	Ph.D. researcher on interpretable reinforcement learning, Inria Scool, Université de Lille, (Advisors: Dr. R. Akrour and Pr. P. Preux)
2021	Research assistant working on unmanned underwater drone control, Lab-STICC, ENSTA, (Advisors: Pr. B. Clement and G. Le Chenadec)

Education

2022-2025	**Ph.D.**, Interpretable Reinforcement Learning, Université de Lille
2020-22	**M.Sc.**, Artificial Intelligence *Excellence Track*, Sorbonne Université
2016-20	**B.Sc.**, Computer Science & Mathematics, University of Manchester

Awards & Honors

2025	Université de Lille International Mobility Grant (1650 €)
2022	AI_PhD Lille (Competitive admission to fully funded Ph.D. program)
2022	Sorbonne Université Computer Science Excellence diploma (1̃0/400 students)

Publications

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Journal Articles

J1. **Hector Kohler**, Akrour, R. & Preux, P. Limits of reinforcement learning for decision trees in Markov decision processes. *Reinforcement Learning Journal* **7** (2026).

J2. Mathieu, T., Medeiros Centa, M., Della Vecchia, R., **Kohler, Hector**, Shilova, A., Maillard Maillard, O.-A. & Preux, P. AdaStop: adaptive statistical testing for sound comparisons of Deep RL agents. *Transactions on Machine Learning Research* (2024).

J3. **Kohler, Hector**, Clement, B., Chaffre, T. & Le Chenadec, G. PID Tuning using Cross-Entropy Deep Learning: a Lyapunov Stability Analysis. *IFAC-PapersOnLine* (2022).

Peer-reviewed Conference Proceedings

C1. **Kohler, Hector**, Akrour, R. & Preux, P. *Breiman meets Bellman: Non-Greedy Decision Trees with MDPs* in *Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining* (2025).

Posters

P1. Mathieu, T., Achddou, J., Davey, A., **Kohler, Hector**, Preux, P. & Teigny, J. *Is Deep RL Reproducible? State of the art and new tools* June 2026. https://inria.hal.science/hal-05645819.

P2. **Kohler, Hector**, Delfosse, Q., Radji, W., Akrour, R. & Preux, P. *Evaluating Interpretable Reinforcement Learning by Distilling Policies into Programs* Workshop on Programmatic Reinforcement Learning (Edmonton, Canada). 2025.

P3. **Kohler, Hector**, Delfosse, Q., Akrour, R., Kersting, K. & Preux, P. *Interpretable and editable programmatic tree policies for reinforcement learning* 17th European Workshop on Reinforcement Learning (Toulouse, France). 2024.

Tools & Software

Core contributor to `rlberry`: PyTorch implementation of deep reinforcement learning agents. Compatibility with `stable-baselines3`. Advanced training analysis with robust stats. Usage of git for development and continuous integration. Organized hackhatons.

Maintainer of a set of fully-transparent control policies for games and robots: inspired by the `stable-baselines3-zoo` and the `gymnasium` API.

Designed statistical test for the reliable and cheap comparison and benchmarking of RL.

Maintainer of my own `scikit-learn` estimator: I developped DPDT, a decision tree induction algorithm which is SOTA for tabular problems.

Teaching

Ecole Normale Supérieure de Cachan (MVA)

2024 Assistant Instructor, Reinforcement Learning (6 hours)

Centrale Lille

2024 Assistant Instructor, Sequential Decision Making (6 hours)

Université de Lille

2023-2024 Assistant Instructor, Algorithms and Programming (60 hours)
2022 Assistant Instructor, Reinforcement Learning (24 hours)

Academic Advising

Graduate

2023	Veronika Shilova, Ecole Polytechnique, Entropy-regularized RL
2023	Quentin Dinel, Université de Lille, Image Segmentation
2023	Mathieu Medeng Essia, Centrale Lille, Imitation Learning
2023	Antonio Al-Makdissi, Centrale Lille, Object-centric Atari Learning
2023	Gabriele Maggioni, Centrale Lille, Model-Based Reinforcement Learning

Academic Service

Reviewing

2027	Senior reviewing for the Reinforcement Learning Journal
2026	RLC, TMLR, ICLR
2025	ICLR, RLC, AISTATS, ICML
2024	NeurIPS

Talks

2026	DaSciM seminars.
2025	Seminar at ENSTA Paris for Sao MAI NGUYEN group.
2025	RL seminar at the University of Alberta in Adam White group.
2025	Orals at RLC and KDD.
2023- 2024	Scool seminars.

Chairing

2026	Program chair in charge of OpenReview for EWRL 2026.
2024	Organized the InterpPol workshop at the first Reinforcement Learning Conference in Amherst (USA).

Responsabilities

2023-2025	Elected representative of Ph.D. students and Postdocs at the CRIStAL laboratory (Université de Lille).
2025	Ph.D. representative for the HCERES evaluation of CRIStAL.

Last updated: November 2, 2026